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AIFBA EXPERIMENT NO : 08

CODE:

```
# ==============================
# STEP 1: Import Libraries
# ==============================
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
# For better visuals
sns.set(style="whitegrid")
# ==============================
# STEP 2: Load Dataset
# ==============================
# Use Kaggle dataset: creditcard.csv
# Upload in Colab before running
url = "https://storage.googleapis.com/download.tensorflow.org/data/creditcard.csv"
df = pd.read_csv(url)
df.head()
print("Dataset Shape:", df.shape)
print(df.head())
# ==============================
# STEP 3: Data Overview
# ==============================
print("\nClass Distribution:")
print(df['Class'].value_counts())
# ==============================
# STEP 4: Fraud vs Non-Fraud Count
# ==============================
plt.figure(figsize=(6,4))
```

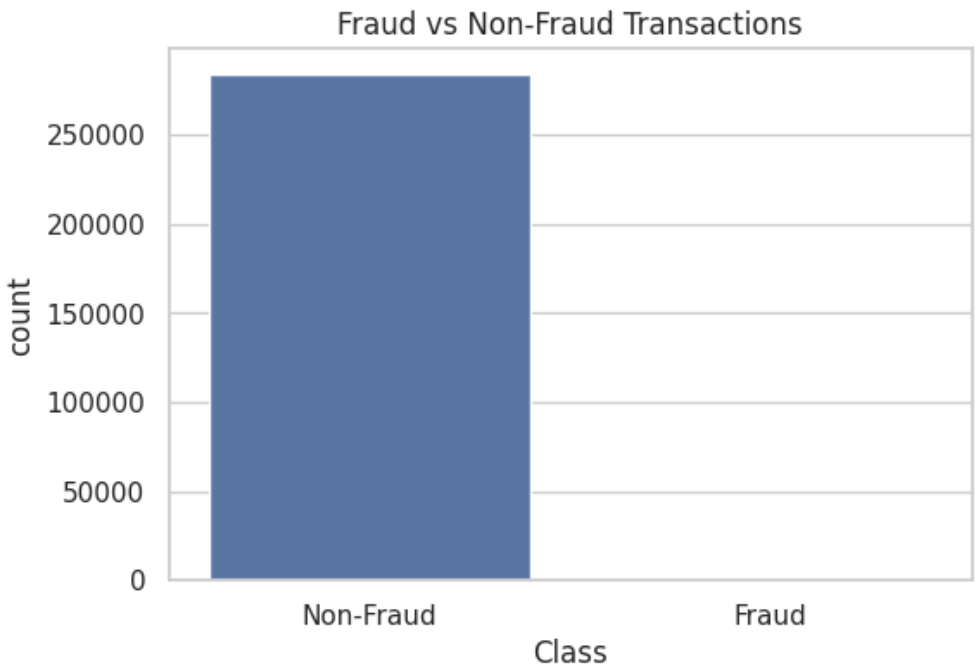
```
sns.countplot(x='Class', data=df)
plt.title("Fraud vs Non-Fraud Transactions")
plt.xticks([0,1], ['Non-Fraud', 'Fraud'])
plt.show()
# ================================
# STEP 5: Transaction Amount Distribution
# ================================
plt.figure(figsize=(8,5))
sns.histplot(df['Amount'], bins=50, kde=True)
plt.title("Transaction Amount Distribution")
plt.show()
# ================================
# STEP 6: Fraud Amount Distribution
# ================================
plt.figure(figsize=(8,5))
sns.boxplot(x='Class', y='Amount', data=df)
plt.title("Amount Distribution by Class")
plt.xticks([0,1], ['Non-Fraud', 'Fraud'])
plt.show()
# ================================
# STEP 7: Correlation Heatmap
# ================================
plt.figure(figsize=(12,10))
corr = df.corr()
sns.heatmap(corr, cmap='coolwarm')
plt.title("Feature Correlation Heatmap")
plt.show()
# ================================
# STEP 8: Time vs Fraud Pattern
# ================================
plt.figure(figsize=(8,5))
sns.histplot(df[df['Class']==1]['Time'], bins=50, color='red')
```

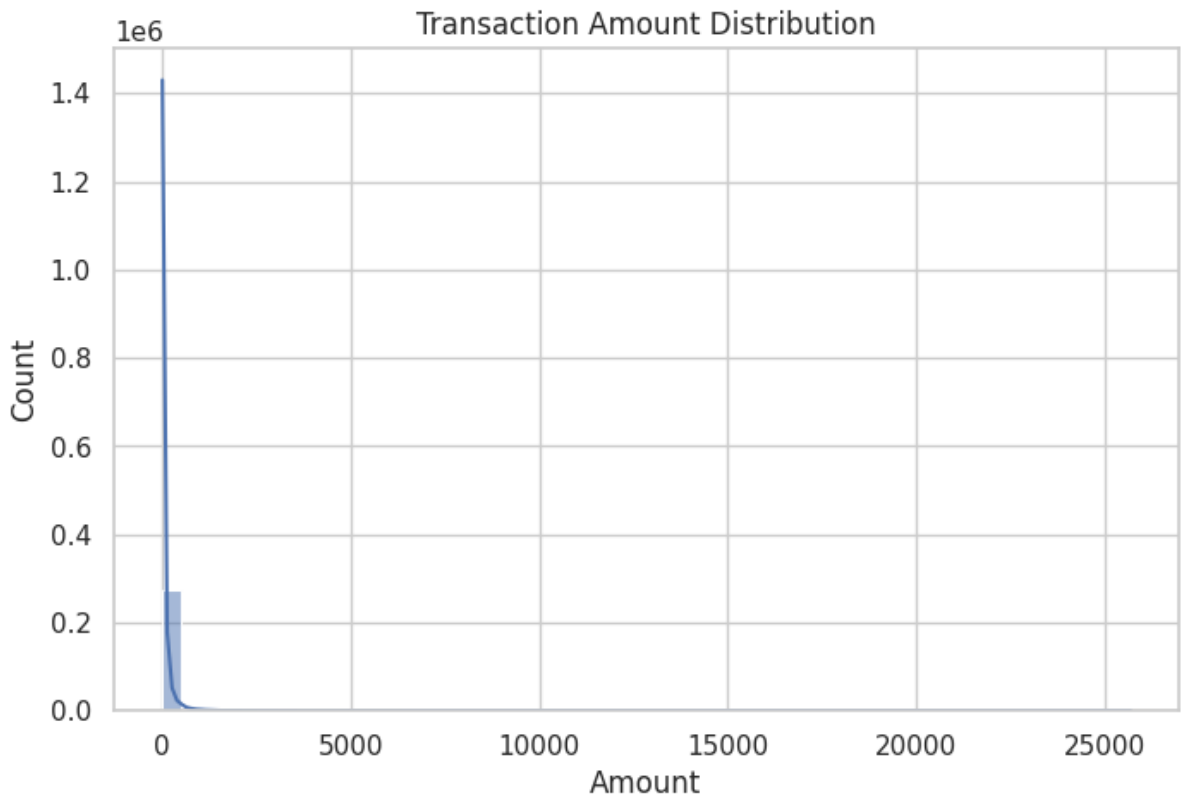
```
plt.title("Fraud Transactions Over Time")
plt.show()
# ================================
# STEP 9: Interactive Plot (Plotly)
# ================================
fig = px.scatter(df.sample(2000),
                 x='Amount',
                 y='Time',
                 color='Class',
                 title="Fraud Detection Scatter Plot")
fig.show()
# ================================
# STEP 10: Fraud Percentage
# ================================
fraud_percentage = (df['Class'].sum() / len(df)) * 100
print(f"\nFraud Transactions Percentage: {fraud_percentage:.4f}%"
# ================================
# STEP 11: Pairplot (Important Features)
# ================================
sample_df = df[['V1', 'V2', 'V3', 'Amount', 'Class']].sample(1000)
sns.pairplot(sample_df, hue='Class')
plt.show()
```

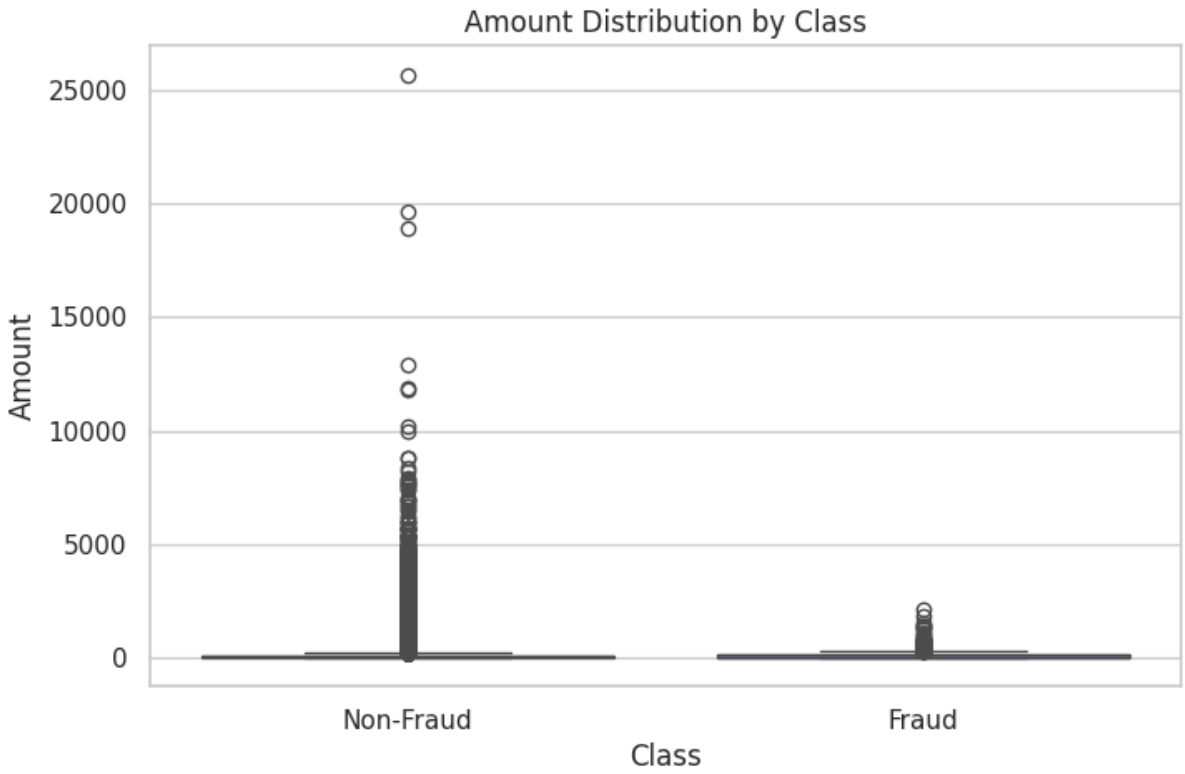
OUTPUT:

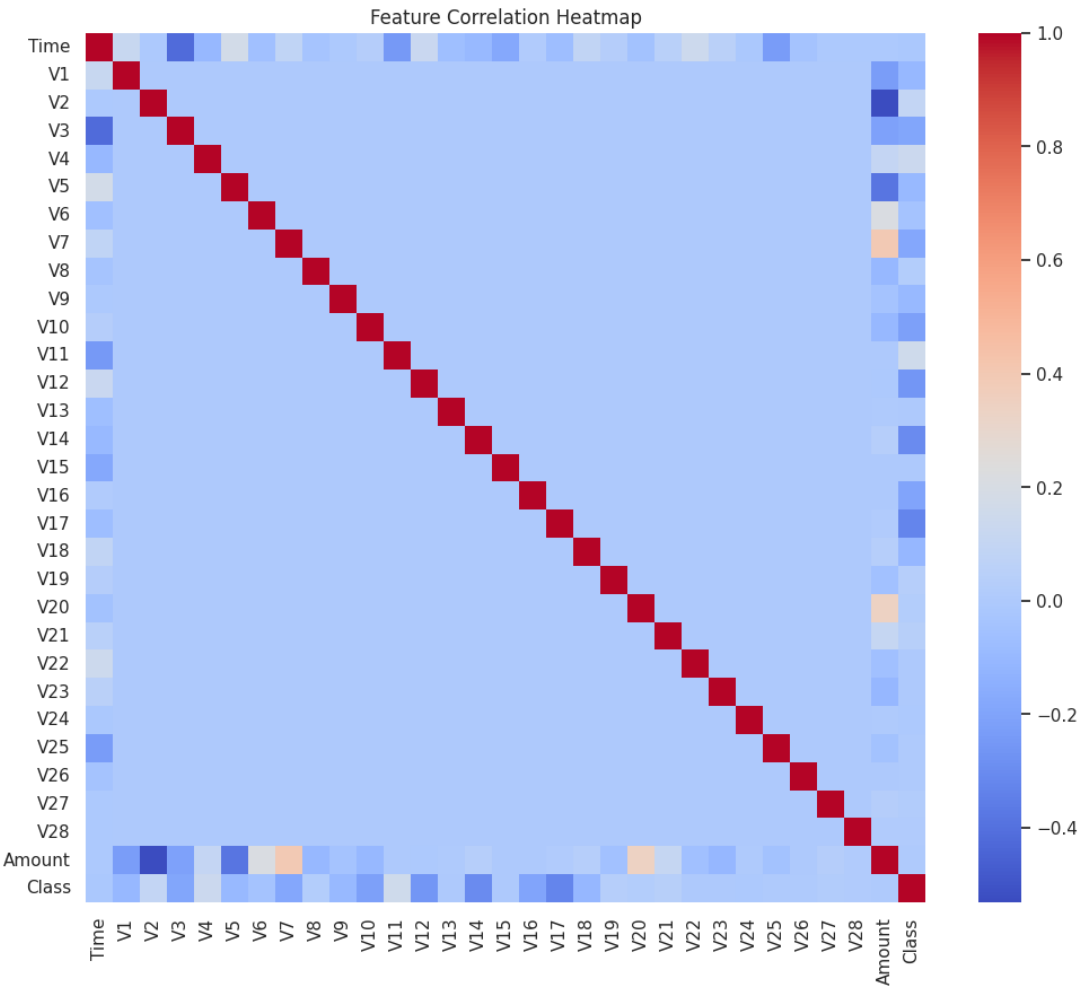
```
Dataset Shape: (284807, 31)
   Time        V1        V2        V3        V4        V5        V6        V7  \
0   0.0 -1.359807 -0.072781  2.536347  1.378155 -0.338321  0.462388  0.239599
1   0.0  1.191857  0.266151  0.166480  0.448154  0.060018 -0.082361 -0.078803
2   1.0 -1.358354 -1.340163  1.773209  0.379780 -0.503198  1.800499  0.791461
3   1.0 -0.966272 -0.185226  1.792993 -0.863291 -0.010309  1.247203  0.237609
4   2.0 -1.158233  0.877737  1.548718  0.403034 -0.407193  0.095921  0.592941

        V8        V9  ...       V21       V22       V23       V24       V25  \
```

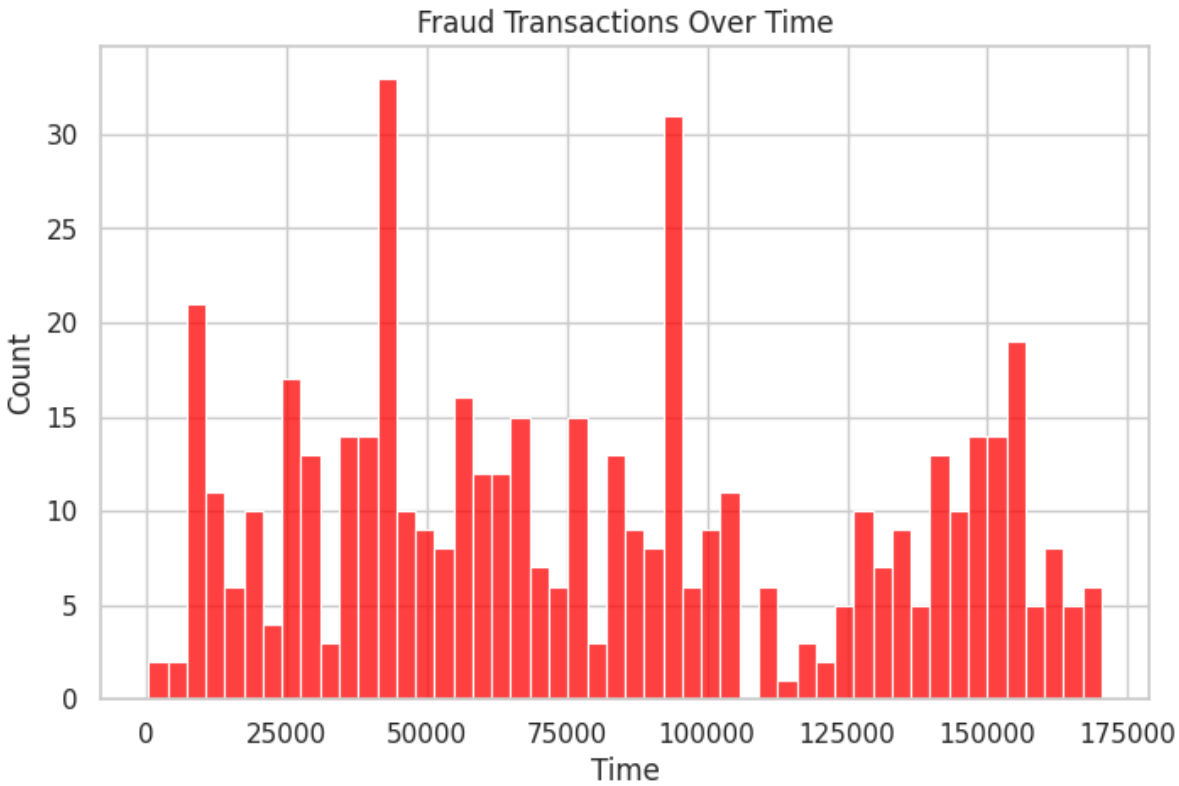
```
0  0.098698  0.363787  ... -0.018307  0.277838 -0.110474  0.066928  0.128539
1  0.085102 -0.255425  ... -0.225775 -0.638672  0.101288 -0.339846  0.167170
2  0.247676 -1.514654  ...  0.247998  0.771679  0.909412 -0.689281 -0.327642
3  0.377436 -1.387024  ... -0.108300  0.005274 -0.190321 -1.175575  0.647376
4 -0.270533  0.817739  ... -0.009431  0.798278 -0.137458  0.141267 -0.206010

       V26       V27       V28  Amount  Class
0 -0.189115  0.133558 -0.021053  149.62      0
1  0.125895 -0.008983  0.014724    2.69      0
2 -0.139097 -0.055353 -0.059752  378.66      0
3 -0.221929  0.062723  0.061458  123.50      0
4  0.502292  0.219422  0.215153   69.99      0

[5 rows x 31 columns]

Class Distribution:
Class
0    284315
1       492
Name: count, dtype: int64
```



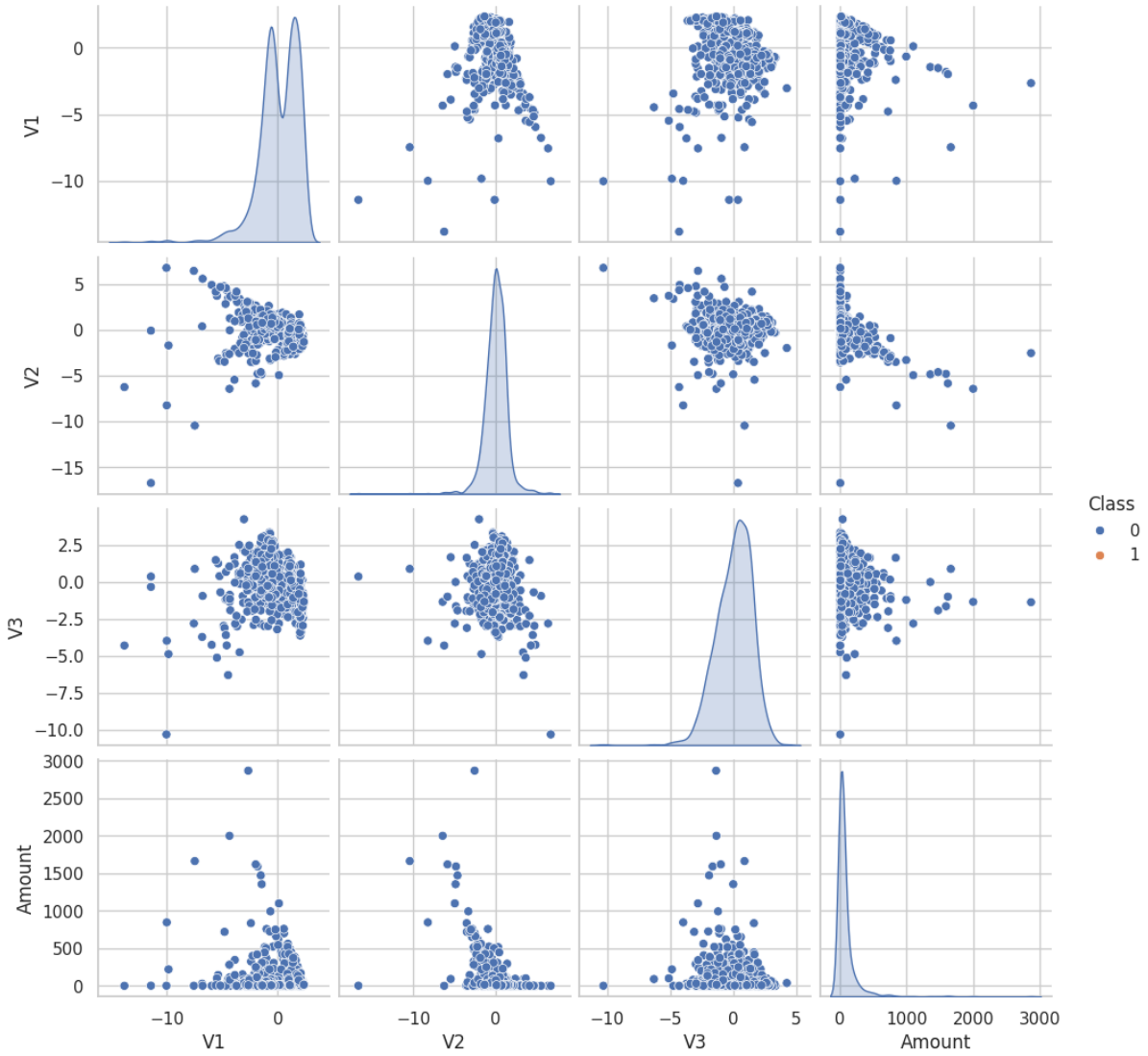

Transaction Amount Distribution
1e6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Count
0
5000
10000
15000
20000
25000
Amount


Amount Distribution by Class
25000
20000
15000
10000
5000
0
Amount
Non-Fraud
Fraud
Class


Feature Correlation Heatmap
Time
V1
V2
V3
V4
V5
V6
V7
V8
V9
V10
V11
V12
V13
V14
V15
V16
V17
V18
V19
V20
V21
V22
V23
V24
V25
V26
V27
V28
Amount
Class
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4


Fraud Transactions Over Time
Count
Time
0
5
10
15
20
25
30
0
25000
50000
75000
100000
125000
150000
175000

02k4k6k8k10k12k050k100k150k

00.20.40.60.81ClassFraud Detection Scatter PlotAmountTime

Fraud Transactions Percentage: 0.1727%